Binomial Theorem

- **Definition:** The number of k-combinations of a set with n elements, denoted by $C(n,k)$ or $\binom{n}{k}$, is also called a **binomial coefficient** because it occurs as a coefficient in the expansion of the power of a binomial expression such as $(x+y)^n$.
- **Theorem (Binomial theorem)**

Let x and y be variables and n be a nonnegative integer. Then

$$(x+y)^n$$

$$= \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$$